# Jiwon Lee

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## Education

**Western University** — London, ON
*BSc Computer Science* — *Sep 2023 – May 2028*

- **Relevant coursework:** Operating Systems, Databases, Machine Learning, Reinforcement Learning, Data Science

## Experience

**Software Engineering Intern** — Sep 2026 – Feb 2027
*Inventis Korea* — *Seoul, South Korea*

- Traced 36 REST API flows through a multi-layer **C#/ASP.NET** dealer management SaaS backend, producing Mermaid sequence diagrams now used to onboard product engineers.
- Analyzed the **VR Builder (Unity) framework** for an XR training-content platform, mapping built-in behaviors and conditions against product requirements to identify custom-development needs, and documented bilingual (Korean/English) design-review decisions and action items for the engineering team.

**Full-Stack Developer** — May 2025 – Aug 2025
*Kingsbridge Rehab Centre* — *Mississauga, ON*

- Built and shipped the clinic's web application end to end with **Next.js 15**, **React 19**, and TypeScript, for a practice handling **1,000+** patient visits.
- Developed a timezone-aware availability module with date-fns that computes open and closed status across the 7-day schedule, eliminating manual updates.

## Projects

**ROCSim** | *C++17, CUDA, HIP/ROCm, Qt 6, OpenGL*

- Delivered a real-time N-body and collision physics engine that runs from **one codebase on both NVIDIA (RTX 3070)** and **AMD (RX 6800) GPUs** at **60 FPS**, by writing portable **CUDA and HIP/ROCm** compute kernels.
- Designed an extensible simulation core where physics models and integrators (Verlet, RK4) are **swappable at runtime without modifying the engine**, using **Strategy and Factory patterns** with **adaptive timestepping** for numerical stability.

**Sentimentrix** | *Python, BERT, scikit-learn, Jupyter*

- Built a Reddit sentiment pipeline that used VADER as **weak supervision** to label **150,000** comments, then trained an SVM on BERT embeddings of 20,000 comments to reproduce those labels at **78.4% accuracy (0.77 weighted F1)**.
- Benchmarked a linear SVM on TF-IDF (unigrams and bigrams) against an RBF SVM on PCA-reduced BERT embeddings, tuning both with GridSearchCV and 3–5 fold cross-validation.

**THOTH** | *Next.js, TypeScript, FastAPI, Supabase (PostgreSQL)*

- **Owned the entire Next.js frontend** for Thoth, a Wi-Fi motion-tracking research platform, integrating a FastAPI backend and Supabase.
- Built the chat interface for the site's AI assistant, letting visitors get answers about the project without contacting the research team.

## Technical Skills

**Languages**: C#, C/C++, Python, TypeScript, JavaScript, SQL (PostgreSQL), Java, HTML/CSS
**Frameworks and Libraries**: ASP.NET, React, React Native, Next.js, Node.js, Express.js, Qt, scikit-learn, pandas, NumPy, Matplotlib, PyTorch, FastAPI
**Tools and Platforms**: Git, Linux, CUDA, HIP/ROCm, OpenGL, Supabase, Jupyter, Google Cloud Platform